

Winard

a window,
controlled by a microcontroller,
programmed by arduino.

By
Wil Straver

Version 0.1

July 2025

Table of Contents

1 Introduction.....	3
2 Windows.....	6
2.1 The primary Window.....	6
2.2 Toplevel window.....	6
3 Widgets.....	7
3.1 containers.....	7
3.1.1 Frame.....	7
3.1.2 Labeled frame.....	7
3.1.3 Notebook.....	8
3.2 Buttons.....	9
3.2.1 button, radiobutton and checkbutton.....	9
3.2.2 menubuttons.....	10
3.2.3 menubar.....	10
3.2.4 separator.....	11
3.3 Entry widgets.....	12
3.3.1 Slider or scale.....	12
3.3.2 Spinbox.....	13
3.3.3 Entry.....	14
3.4 Canvas.....	15
4 Themes and styles.....	16
4.1 Toplevel.....	17
4.2 Text style.....	18
4.3 Buttons style.....	20
4.3.1 button, radiobutton and checkbutton.....	20
4.3.2 Style map.....	21
4.3.3 Button size.....	22
4.4 Frame.....	23
4.5 Notebook.....	24
4.6 Menubutton.....	25
4.7 Menubar.....	26
4.8 Slider or scale.....	27
4.9 Spinbox.....	28
4.10 Entry.....	29
5 Example's.....	30
5.1 Switch leds on-off.....	30
5.2 Led brightness with slider.....	31
5.3 Draw functions.....	32
5.4 Plot an ADC signal.....	34
5.5 Signal a warning.....	35

1 Introduction

Tested with the arduino nano and some with the stm32f103c8

Next page gives an example of how to program a window with content.

code microcontroller

1830 bytes of program storage space

```
void setup( )
{ Serial.begin(115200);
  build_up();
}

void loop( ) {}

void build_up()
{ Serial.println("105,1,");
  Serial.println("2,Arduino Display 1");
  Serial.println("41,0,320,240,");
  Serial.println("2,Build up with Tkinter,");
  Serial.println("24,3,41,0,100,35,");
  Serial.println("2,button");
  Serial.println("25,6,41,0,120,80,");
  Serial.println("2,radiobutton1");
  Serial.println("26,7,41,0,110,120,");
  Serial.println("2,radiobutton2");
  Serial.println("26,8,41,0,110,140,");
  Serial.println("2,checkboxbutton");
  Serial.println("27,9,41,0,110,160,");
  Serial.println("104,1,");
}
```

code, more readable, same result

3448 bytes of program storage space

```
# define TEXT      2
# define LABEL    24
# define BUTTON   25
# define RBUTTON  26
# define CBUTTON  27
# define ROOT     41
# define PRINTOBJECTS 104
# define PRINTBUILD 105

char print_str[64];

void setup( )
{ Serial.begin(115200);
  build_up();
}

void loop( ) {}

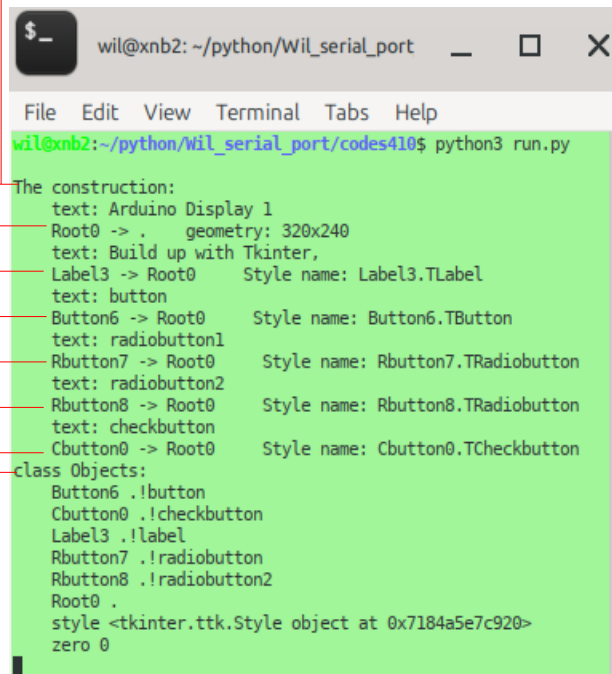
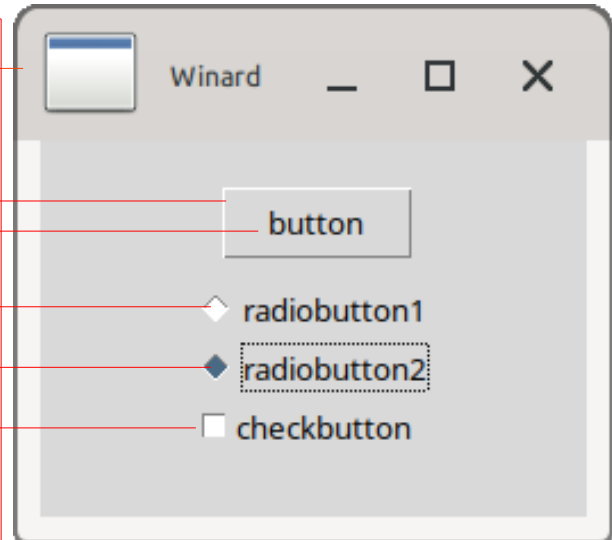
void build_up()
{ tx_02(PRINTBUILD,1);
  tx_str(TEXT,"Arduino Display");
  tx_04(ROOT,0,320,240);
  tx_str(TEXT,"Build up with Tkinter");
  tx_06(LABEL,3,ROOT,0,100,35);
  tx_str(TEXT,"button");
  tx_06(BUTTON,6,ROOT,0,120,80);
  tx_str(TEXT,"radiobutton1");
  tx_06(RBUTTON,7,ROOT,0,110,120);
  tx_str(TEXT,"radiobutton2");
  tx_06(RBUTTON,8,ROOT,0,110,145);
  tx_str(TEXT,"checkboxbutton");
  tx_06(CBUTTON,9,ROOT,0,110,170);
  tx_02(PRINTOBJECTS,1);
}

void tx_02(int a,int b)
{ sprintf(print_str,"%d,%d",a,b);
  Serial.println(print_str);
}

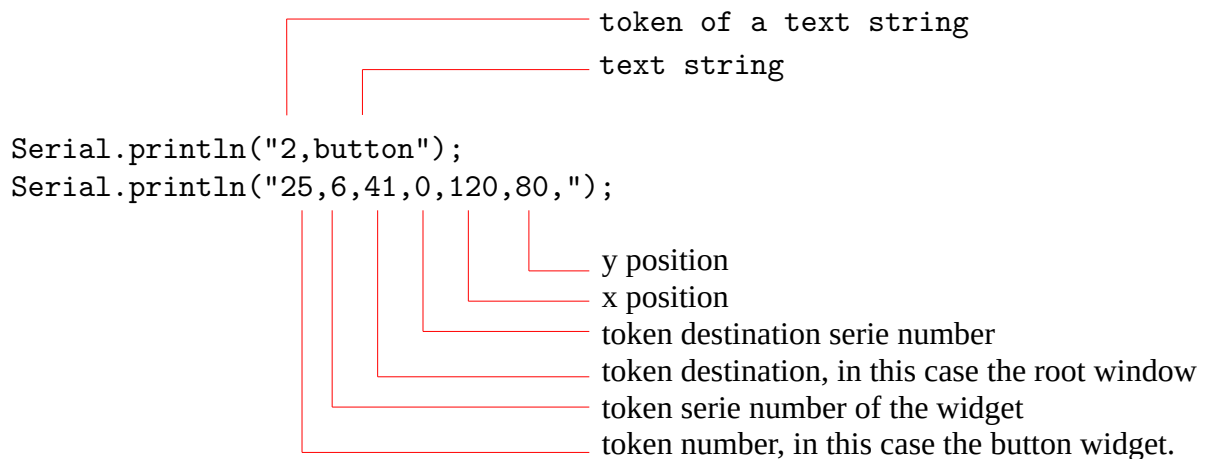
void tx_04(int a,int b,int c,int d)
{ sprintf(print_str,"%d,%d,%d,%d",a,b,c,d);
  Serial.println(print_str);
}

void tx_06(int a,int b,int c,int d,int e,int f)
{ sprintf(print_str,"%d,%d,%d,%d,%d,%d",
  a,b,c,d,e,f);
  Serial.println(print_str);
}

void tx_str(int a,const char *str)
{ sprintf(print_str,"%d,%s",a,str);
  Serial.println(print_str);
}
```



Short explanation of coding, the yellow rectangle:



```
Serial.println("2,"button"");
Serial.println("25,6,41,0,120,80,");
```

In tkinter language:

The first token is the child widget and the second token, the destination, is the parent widget.

When macros are used the code becomes more readable.

The code, in the light blue rectangle:

```
# define TEXT          2
# define BUTTON       25
etc.

tx_str(TEXT,"button");
tx_06(BUTTON,6,ROOT,0,120,80);
etc

void tx_str(int a,const char *str)
{ sprintf(print_str,"%d,%s",a,str);
  Serial.println(print_str);
}

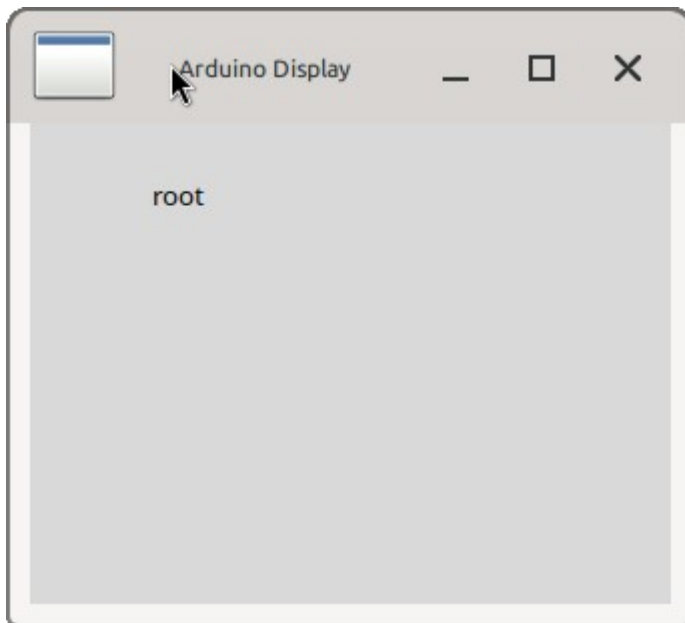
void tx_06(int a,int b,int c,int d,int e,int f)
{ sprintf(print_str,"%d,%d,%d,%d,%d,%d",a,b,c,d,e,f);
  Serial.println(print_str);
}
etc.
```

A collection of macros and functions are added if we include the header file <winard.h>.

From here on, all examples will use the include file <winard.h>.

2 Windows

2.1 The primary Window



```
#include <winard.h>

void setup( )
{ Serial.begin(115200);
  build_up();
}

void loop( ) {;}

void build_up()
{ tx_02(PRINTBUILD,1);

  tx_str(TEXT,"Arduino Display");
  tx_04(ROOT,0,320,240);

  tx_str(TEXT,"root");
  tx_06(LABEL,0,ROOT,0,60,25);

  tx_02(PRINTOBJECTS,1);
}
```

2.2 Toplevel window

Child of the primary window

```
#include <winard.h>

void setup( )
{ Serial.begin(115200);
  build_up();
}

void loop( ) {;}

void build_up()
{ tx_02(PRINTBUILD,1);

  tx_str(TEXT,"Display 0");
  tx_04(ROOT,0,320,240);

  tx_str(TEXT,"Display 1");
  tx_04(TOPLEVEL,1,320,240);

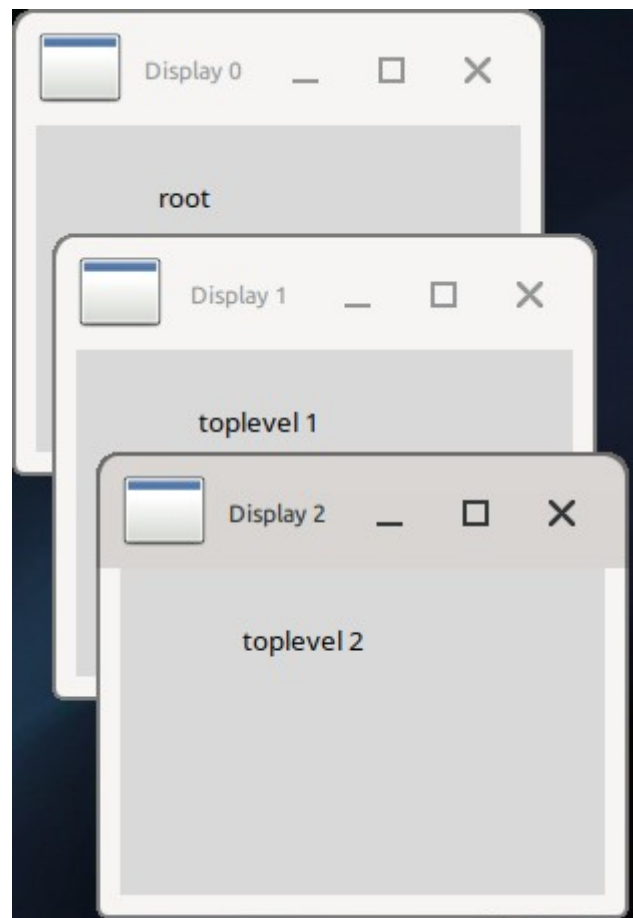
  tx_str(TEXT,"Display 2");
  tx_04(TOPLEVEL,2,320,240);

  tx_str(TEXT,"root");
  tx_06(LABEL,0,ROOT,0,60,25);

  tx_str(TEXT,"toplevel 1");
  tx_06(LABEL,1,TOPLEVEL,1,60,25);

  tx_str(TEXT,"toplevel 2");
  tx_06(LABEL,2,TOPLEVEL,2,60,25);

  tx_02(PRINTOBJECTS,1);
}
```



3 Widgets

3.1 containers

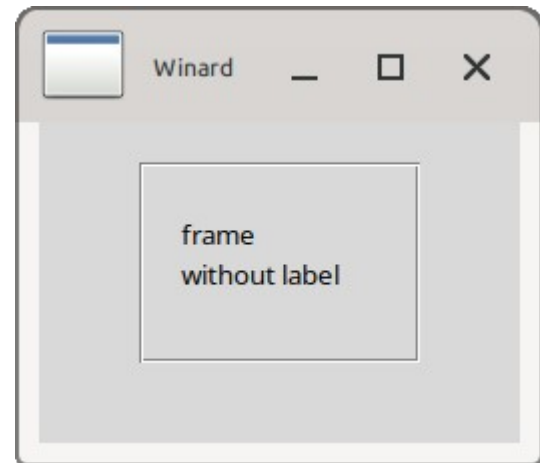
3.1.1 Frame

```
#include <winard.h>

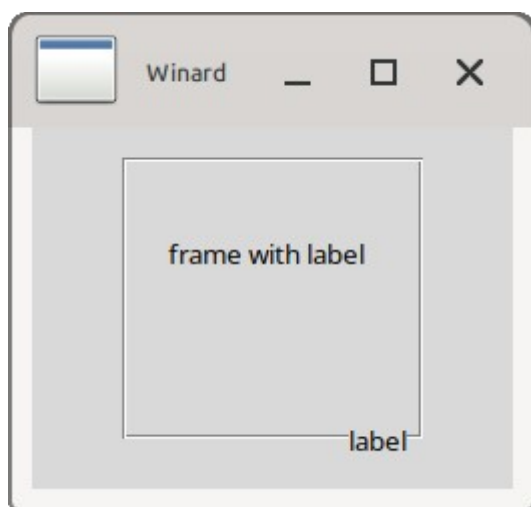
void setup( )
{ Serial.begin(115200);
  build_up();
}

void loop() {}

void build_up()
{ tx_02(PRINTBUILD,1);
  tx_str(TEXT,"Winard");
  tx_04(ROOT,0,240,160);
  tx_08(FRAME,1,ROOT,0,50,20,140,100);
  tx_str(TEXT,"frame");
  tx_06(LABEL,4,FRAME,1,20,25);
  tx_str(TEXT,"without label");
  tx_06(LABEL,5,FRAME,1,20,45);
  tx_02(RELIEF,1);
  tx_05(STYLE,0,FRAME,1,RELIEF);
  tx_02(BORDERWIDTH,2);
  tx_05(STYLE,0,FRAME,1,BORDERWIDTH);
  tx_02(PRINTOBJECTS,1);
}
```



3.1.2 Labeled frame



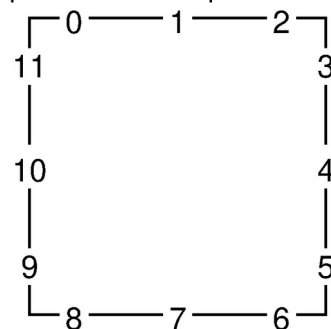
```
#include <winard.h>

void setup( )
{ Serial.begin(115200);
  build_up();
}

void loop() {}

void build_up()
{ tx_02(PRINTBUILD,1);
  tx_str(TEXT,"Winard");
  tx_04(ROOT,0,240,180);
  tx_str(TEXT,"label");
  tx_02(POSITION,6);
  tx_08(LABELFRAME,1,ROOT,0,45,15,150,150);
  tx_str(TEXT,"frame with label");
  tx_06(LABEL,5,LABELFRAME,1,20,35);
  tx_02(PRINTOBJECTS,1);
}
```

possible label positions



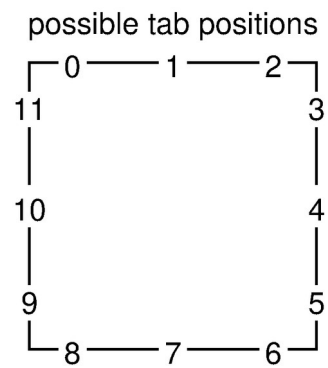
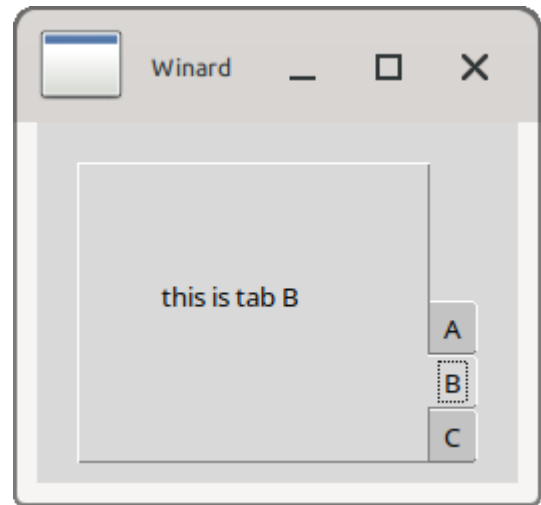
3.1.3 Notebook

```
#include <winard.h>

void setup( )
{ Serial.begin(115200);
  build_up();
}

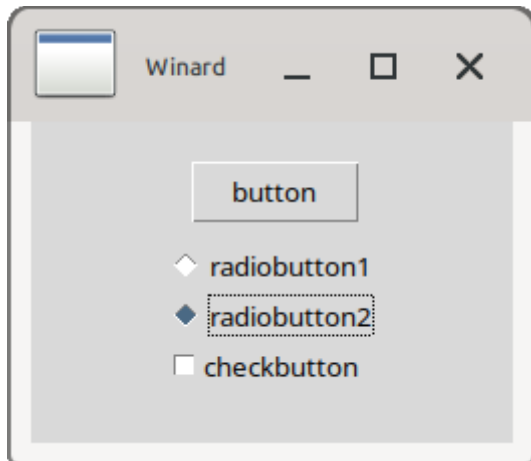
void loop() {;}

void build_up()
{ tx_02(PRINTBUILD,1);
  tx_str(TEXT,"Winard");
  tx_04(ROOT,0,240,180);
  tx_str(TEXT,"A,B,C");
  tx_02(POSITION,5);
  tx_08(NOTEBOOK,1,ROOT,0,20,20,200,150);
  tx_str(TEXT,"this is tab A");
  tx_06(LABEL,4,TAB,0,40,35);
  tx_str(TEXT,"this is tab B");
  tx_06(LABEL,5,TAB,1,40,55);
  tx_str(TEXT,"this is tab C");
  tx_06(LABEL,6,TAB,2,40,75);
  tx_02(PRINTOBJECTS,1);
}
```



3.2 Buttons

3.2.1 button, radiobutton and checkbutton



```
#include <winard.h>

void setup( )
{ Serial.begin(115200);
  build_up();
}

void loop( ) {;}

void build_up()
{ tx_02(PRINTBUILD,1);
  tx_str(TEXT,"Winard");
  tx_04(ROOT,0,240,160);
  tx_str(TEXT,"button");
  tx_06(BUTTON,6,ROOT,0,80,20);
  tx_str(TEXT,"radiobutton1");
  tx_str(TAGS,"rb");
  tx_06(RBUTTON,7,ROOT,0,70,60);
  tx_str(TEXT,"radiobutton2");
  tx_06(RBUTTON,8,ROOT,0,70,85);
  tx_str(TEXT,"checkbutton");
  tx_06(CBUTTON,9,ROOT,0,70,110);
  tx_02(PRINTOBJECTS,1);
}
```

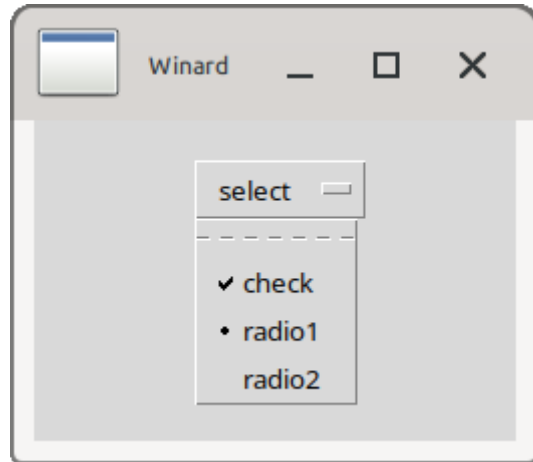
3.2.2 menubuttons

```
#include <winard.h>

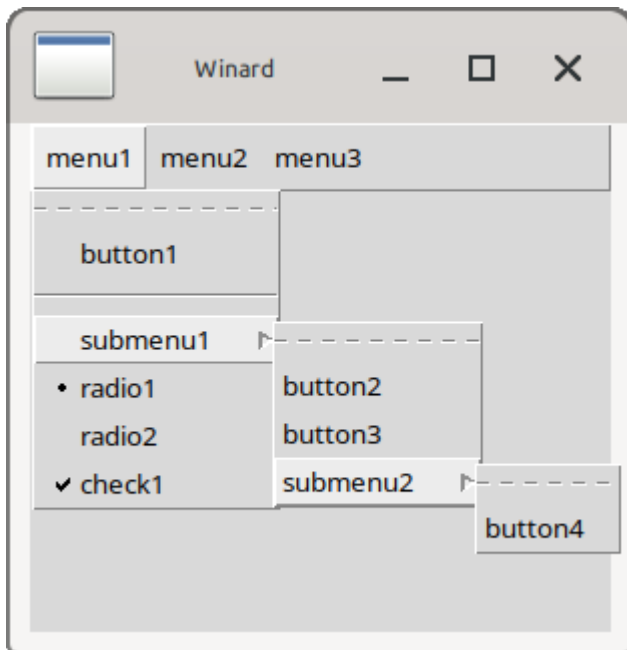
void setup( )
{ Serial.begin(115200);
  the_build();
}

void loop() {}

void the_build()
{ tx_02(PRINTBUILD,1);
  tx_str(TEXT,"Winard");
  tx_04(ROOT,0,240,160);
  tx_str(TEXT,"select");
  tx_06(MBUTTON,2,ROOT,0,80,20);
  tx_str(TEXT,"check");
  tx_04(MCBUTTON,3,MBUTTON,2);
  tx_str(TAGS,"rb");
  tx_str(TEXT,"radio1");
  tx_04(MRBUTTON,5,MBUTTON,2);
  tx_str(TEXT,"radio2");
  tx_04(MRBUTTON,6,MBUTTON,2);
  tx_02(PRINTOBJECTS,1);
}
```



3.2.3 menubar



```
#include <winard.h>
void setup( )
{ Serial.begin(115200);
  build_up();
}

void loop() {}

void build_up()
{ tx_02(PRINTBUILD,1);
  tx_str(TEXT,"Winard");
  tx_04(ROOT,0,290,220);
  tx_02(TEAR,1);
  tx_str(TEXT,"menu1,menu2,menu3");
  tx_04(MENUBAR,0,ROOT,0);
  tx_str(TEXT,"button1");
  tx_04(BMCOMMAND,1,BARMENU,0);
  tx_04(BMSEPARATOR,2,BARMENU,0);
  tx_str(TEXT,"submenu1");
  tx_04(SUBMENU,0,BARMENU,0);
  tx_str(TEXT,"button2");
  tx_04(BMCOMMAND,2,SUBMENU,0);
  tx_str(TEXT,"button3");
  tx_04(BMCOMMAND,3,SUBMENU,0);
  tx_str(TEXT,"submenu2");
  tx_04(SUBMENU,2,SUBMENU,0);
  tx_str(TEXT,"button4");
  tx_04(BMCOMMAND,4,SUBMENU,2);
  tx_str(TAGS,"rb2");
  //TAGS, connects the radio buttons
  // with this label
  tx_str(TEXT,"radio1");
  tx_04(BMRADIO,1,BARMENU,0);
  tx_str(TEXT,"radio2");
  tx_04(BMRADIO,2,BARMENU,0);
  tx_str(TEXT,"check1");
  tx_04(BMCHECK,1,BARMENU,0);
  tx_str(TEXT,"button5");
  tx_04(BMCOMMAND,5,BARMENU,1);
  tx_str(TEXT,"button6");
  tx_04(BMCOMMAND,6,BARMENU,1);
  tx_04(BMSEPARATOR,0,BARMENU,1);
  tx_str(TEXT,"button7");
  tx_04(BMCOMMAND,7,BARMENU,1);
  tx_02(PRINTOBJECTS,1);
}
```

3.2.4 separator

With separator you can draw a vertical or horizontal line.

```
#include <winard.h>

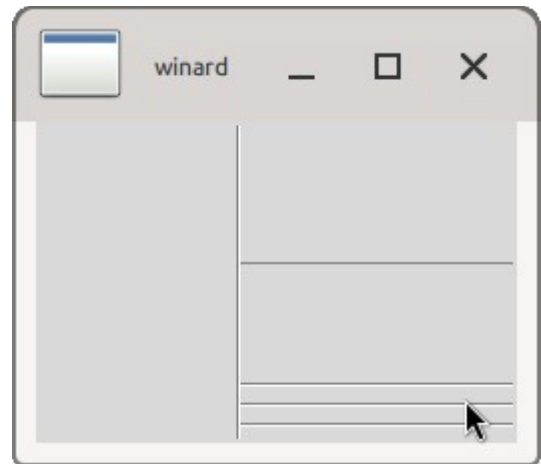
void setup( )
{ Serial.begin(115200);
  build_up();
}

void loop( ) {}

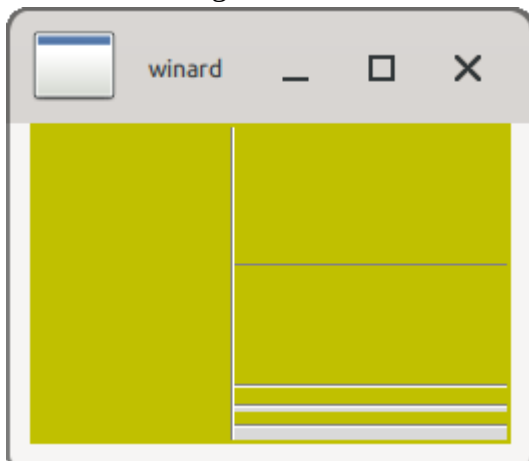
void build_up()
{ tx_02(PRINTBUILD,1);
  tx_str(TEXT,"winard");
  tx_04(ROOT,0,240,160);

  tx_08(SEPARATOR,1,ROOT,0,100,70,138,1);
  tx_08(SEPARATOR,1,ROOT,0,100,130,138,2);
  tx_08(SEPARATOR,1,ROOT,0,100,140,138,4);
  tx_08(SEPARATOR,1,ROOT,0,100,150,138,8);
  tx_08(SEPARATOR,2,ROOT,0,100,2,2,156);

  tx_02(PRINTOBJECTS,1);
}
```



To see better the line width,
use an other background color:



```
#include <winard.h>

void setup( )
{ Serial.begin(115200);
  build_up();
}

void loop( ) {}

void build_up()
{ tx_02(PRINTBUILD,1);
  tx_str(TEXT,"winard");
  tx_04(ROOT,0,240,160);

  tx_04(CBG,YELLOW);
  tx_05(STYLE,0,ROOT,0,CBG);

  tx_08(SEPARATOR,1,ROOT,0,100,70,138,1);
  tx_08(SEPARATOR,1,ROOT,0,100,130,138,2);
  tx_08(SEPARATOR,1,ROOT,0,100,140,138,4);
  tx_08(SEPARATOR,1,ROOT,0,100,150,138,8);
  tx_08(SEPARATOR,2,ROOT,0,100,2,2,156);

  tx_02(PRINTOBJECTS,1);
}
```

Horizontal line:

```
tx_10(SEPARATOR,1,ROOT,0,100,70,138,1);
```

Diagram illustrating the parameters of the `tx_10` function call for a horizontal line:

- separator number
- destination
- xstart
- y start
- length
- width

Vertical line:

```
tx_10(SEPARATOR,2,ROOT,0,100,2,2,156);
```

Diagram illustrating the parameters of the `tx_10` function call for a vertical line:

- length
- width

3.3 Entry widgets

3.3.1 Slider or scale

```
#include <winard.h>

void setup( )
{ Serial.begin(115200);
  build_up();
}

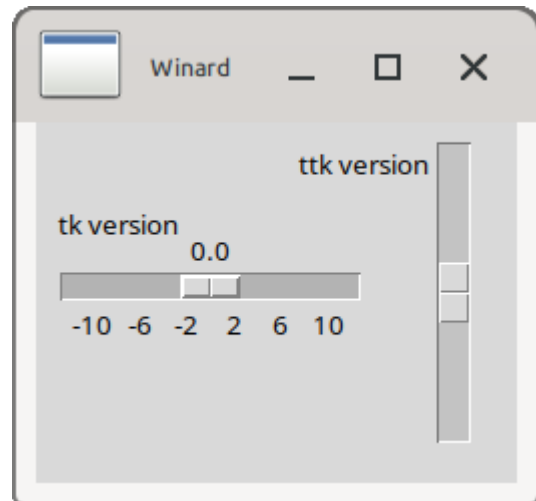
void loop() {;}

void build_up()
{ tx_02(PRINTBUILD,1);
  tx_str(TEXT,"Winard");
  tx_04(ROOT,0,240,180);

  tx_str(TEXT,"0 1 4.0 -10.0 10.0 2.0 1");
  tx_08(SCALE,18,ROOT,0,10,50,150,12);
  tx_str(TEXT,"tk version");
  tx_06(LABEL,1,ROOT,0,10,40);

  tx_str(TEXT,"1 1 2.0 20.0 -20.0 2.0 2");
  tx_08(SCALE,19,ROOT,0,200,10,0,150);
  tx_str(TEXT,"ttk version");
  tx_06(LABEL,1,ROOT,0,130,10);

  tx_02(PRINTOBJECTS,1);
}
```



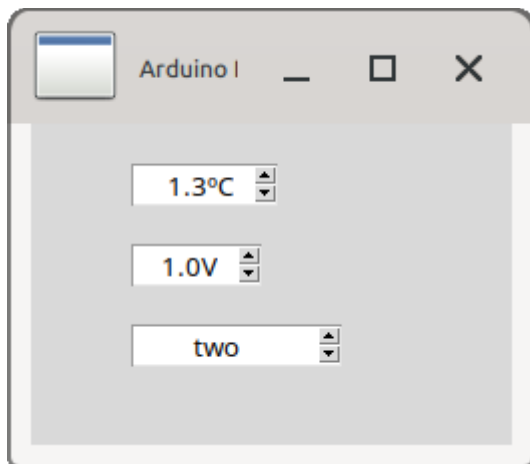
```
tx_str(TEXT,"0 1 4.0 -10.0 10.0 2.0 1");
```

resolution 1=10⁻¹ 2=10⁻² etc.
set
to
from
interval
showvalue
type 0=tk 1=ttk

```
tx_08(SCALE,18,ROOT,0,10,50,150,12);
```

Y2,(if Y2 < X2) Y2=width //vertical
else: Y2=length
X2,(if X2 < Y2) X2=width //horizontal
else: X2=length
Y1 start coord.
X1 start coord.

3.3.2 Spinbox



```
#include <winard.h>

void setup( )
{ Serial.begin(115200);
  build_up();
}

void loop() {;}

void build_up()
{ tx_02(PRINTBUILD,1);
  tx_str(TEXT,"Arduino Display");
  tx_04(ROOT,0,240,160);

  tx_str(SPINBOXSET,"0 2 6 1.0°C");
  tx_str(TEXT,"0.0 10.0 0.1 %.1f°C ");
  tx_06(SPINBOX,1,ROOT,0,50,20);

  // remember, next line not longer then the
  // buffer length, including TEXT, spaces
  // and the two newline/feedback characters

  tx_str(TEXT,"-0.2V -0.1V 0.0V 0.1V 0.2V
              0.5V 1.0V");
  tx_str(SPINBOXSET,"1 2 5 0.2V");
  tx_06(SPINBOX,2,ROOT,0,50,60);

  tx_str(SPINBOXSET,"1 1 10 four");
  tx_str(TEXT,"one two three four five");
  tx_06(SPINBOX,3,ROOT,0,50,100);
}
```

```
tx_str(SPINBOXSET,"0 2 6 1.0°C");
```

— 0 display type
 use TEXT:
 tx_str(TEXT,"0.0 10.0 0.1 %.1f°C ");
 — format
 — increment
 — to
 — from

— 1 display type
 use TEXT:
 tx_str(TEXT,"0.2 0.4 0.6 0.8 1.0 1.2")

3.3.3 Entry

```
#include <winard.h>

void setup( )
{ Serial.begin(115200);
  build_up();
}

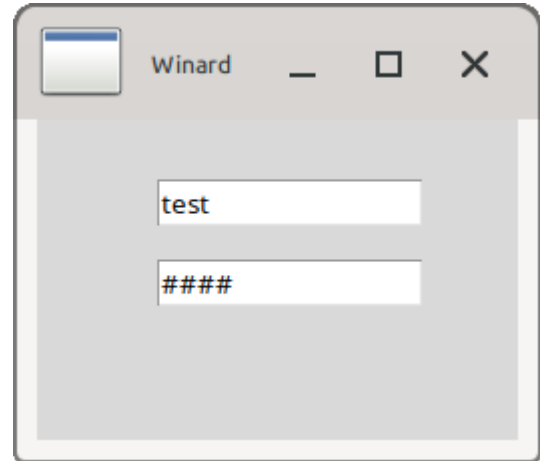
void loop() {;}

void build_up()
{ tx_02(PRINTBUILD,1);
  tx_str(TEXT,"Winard");
  tx_04(ROOT,0,240,160);

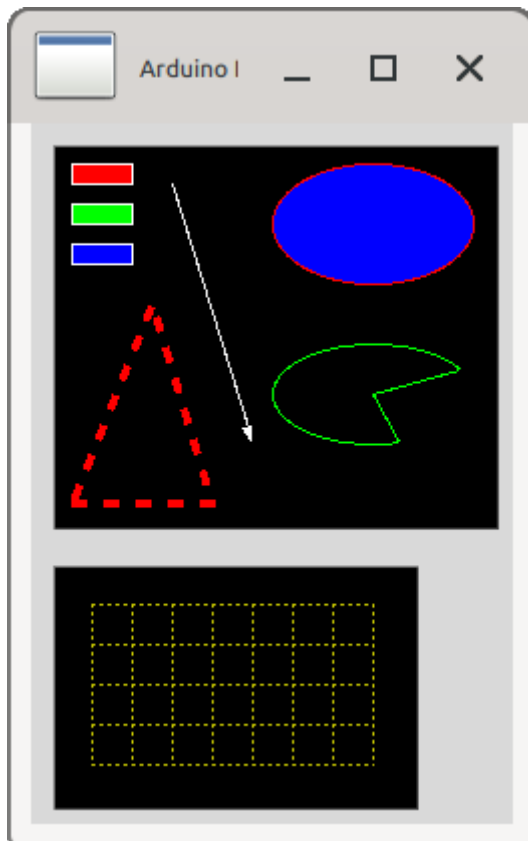
  tx_02(WIDTH,16);
  tx_06(ENTRY,3,ROOT,0,60,30);

  tx_str(SHOW,"#");
  tx_06(ENTRY,14,ROOT,0,60,70);

  tx_02(PRINTOBJECTS,1);
}
```



3.4 Canvas



```
#include <winard.h>

void setup( )
{ Serial.begin(115200);      //(115200);
  build_up();
}

void loop()
{ delay(500) ;
  tx_02(CANVASSELECT,0);
  tx_str(TAGSDEL,"rect");
  delay(500) ;
  rectangles();
}

void build_up()
{ tx_02(PRINTBUILD,1);
  tx_str(TEXT,"Arduino Display 0");
  tx_04(ROOT,0,240,350);

  tx_04(CFG,255,255,255);
  tx_04(CBG,0,0,0);
  tx_08(CANVAS,0,ROOT,0,10,10,220,190);

  rectangles();
  tx_str(TAGS,"line");
  tx_02(LINEWIDTH,1);
  tx_02(ARROW,2);
  tx_04(ARROW,14,20,8);
  tx_08(LINE,5,CANVAS,0,60,20,100,150);

  tx_04(DASH,1,8,8);
  tx_04(CFG,255,0,0);
  tx_02(ARROW,0);
  tx_02(LINEWIDTH,4);
  tx_12(LINE,6,CANVAS,0,10,180,80,180,50,80,10,180);
  tx_04(DASH,0,8,8);
  tx_02(LINEWIDTH,1);
  tx_04(CBG,0,0,255);
  tx_08(OVAL,1,CANVAS,0,110,10,210,70);
  // the oval is drawn in a rectangle X1,Y1  X2,Y2
  // draw a circle: X2-X1=Y2-Y1
  tx_04(CBG,0,0,0);
  tx_04(CFG,0,255,0);
  tx_04(ARCShape,30,255,1);
  //          | | |_0=arc 1=pieslice 2=chord
  //          | | |___extend corner
  //          | | |___start corner
  tx_08(ARC,1,CANVAS,0,110,100,210,150);
  // the underground shape of the arc is just like the
  oval

  tx_04(CBG,0,0,0);
  tx_08(CANVAS,1,ROOT,0,10,220,180,120);
  tx_05(NLINES,5,8,20,20);
  tx_04(CFG,200,200,0);
  tx_02(LINEWIDTH,1);
  tx_04(DASH,1,2,2);
  tx_08(VLINES,3,CANVAS,1,20,20,20,100);
  tx_08(HLINES,3,CANVAS,1,20,20,161,20);
  tx_04(DASH,0,2,2);
  // rectangles();

  tx_02(PRINTOBJECTS,1);
  tx_02(PRINTBUILD,0);
}

void rectangles()
{ tx_str(TAGS,"rect");
  tx_04(CFG,255,255,255);
  tx_04(CBG,255,0,0);
  tx_08(RECT,5,CANVAS,0,10,10,40,20);
  tx_04(CBG,0,255,0);
  tx_08(RECT,6,CANVAS,0,10,30,40,40);
  tx_04(CBG,0,0,255);
  tx_08(RECT,7,CANVAS,0,10,50,40,60);
}
```

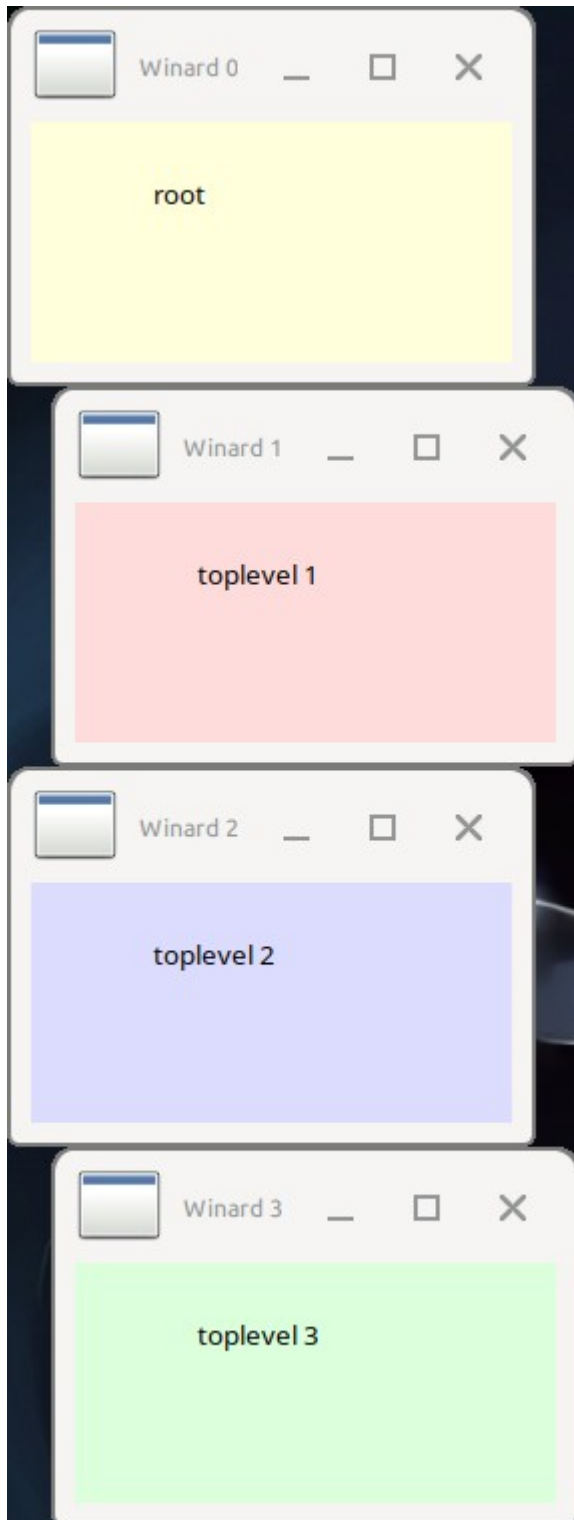
4 Themes and styles

Themes is a collection of styles.

In a theme, all widgets have a default style. It is possible to change the style of a particular widget, for example a button, individually or all of them.

When the style of a widget is changed and then the theme is changed, the style of the new theme is visible, not the changed style.

4.1 Toplevel



```
#include <winard.h>

void setup( )
{ Serial.begin(115200);
  build_up();
}

void loop( ) {};

void build_up()
{ tx_02(PRINTBUILD,1);
  int x=240, y=120;

  tx_str(TEXT,"Winard 0");
  tx_04(ROOT,0,x,y);

  tx_str(TEXT,"Winard 1");
  tx_04(TOPLEVEL,1,x,y);

  tx_str(TEXT,"Winard 2");
  tx_04(TOPLEVEL,2,x,y);

  tx_str(TEXT,"Winard 3");
  tx_04(TOPLEVEL,3,x,y);

  tx_str(TEXT,"root");
  tx_06(LABEL,0,ROOT,0,60,25);

  tx_str(TEXT,"toplevel 1");
  tx_06(LABEL,1,TOPLEVEL,1,60,25);

  tx_str(TEXT,"toplevel 2");
  tx_06(LABEL,2,TOPLEVEL,2,60,25);

  tx_str(TEXT,"toplevel 3");
  tx_06(LABEL,3,TOPLEVEL,3,60,25);

  tx_04(CBG,255,255,220);
  tx_05(STYLE,0,ROOT,0,CBG);
  tx_05(STYLE,0,LABEL,0,CBG);

  tx_04(CBG,255,220,220);
  tx_05(STYLE,0,TOPLEVEL,1,CBG);
  tx_05(STYLE,0,LABEL,1,CBG);

  tx_04(CBG,220,220,255);
  tx_05(STYLE,0,TOPLEVEL,2,CBG);
  tx_05(STYLE,0,LABEL,2,CBG);

  tx_04(CBG,220,255,220);
  tx_05(STYLE,0,TOPLEVEL,3,CBG);
  tx_05(STYLE,0,LABEL,3,CBG);

  tx_02(PRINTOBJECTS,1);
}
```

4.2 Text style

```
tx_10(FONT,0,BUTTON,6,2,16,0,1,0,0);
```

strike
 underlined
 italic
 bold
 size
 fonttype 0 Helvetica 1 Times 2 Arial 3 Verdana
 destination
 nothing here, but used in the case of labelframe and notebook.
 font style

```
#include <winard.h>

void setup( )
{ Serial.begin(115200);
  build_up();
}

void loop( ) {}

void build_up()
{ tx_02(PRINTBUILD,1);
  tx_str(TEXT,"Winard");
  tx_04(ROOT,0,240,160);

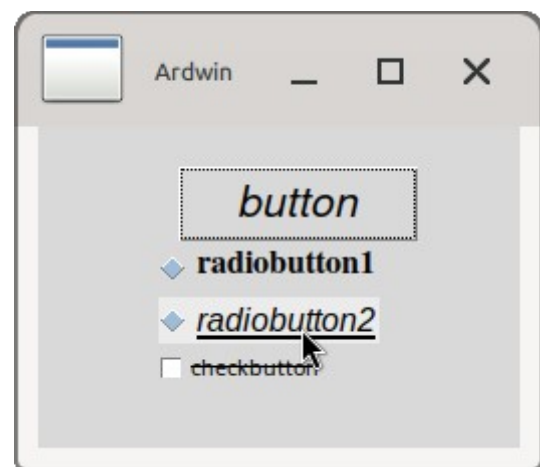
  tx_str(TEXT,"button");
  tx_06(BUTTON,6,ROOT,0,70,20);
  tx_10(FONT,0,BUTTON,6,0,16,0,1,0,0);

  tx_str(TAGS,"rb");
  tx_str(TEXT,"radiobutton1");
  tx_06(RBUTTON,7,ROOT,0,60,60);
  tx_10(FONT,0,RBUTTON,7,1,12,1,0,0,0);

  tx_str(TEXT,"radiobutton2");
  tx_06(RBUTTON,8,ROOT,0,60,85);
  tx_10(FONT,0,RBUTTON,8,2,12,0,1,1,0);

  tx_str(TEXT,"checkbutton");
  tx_06(CBUTTON,9,ROOT,0,60,110);
  tx_10(FONT,0,CBUTTON,9,3,8,0,0,0,1);

  tx_02(PRINTOBJECTS,1);
}
```



```
#include <winard.h>

void setup()
{ Serial.begin(115200);
  build_up();
}

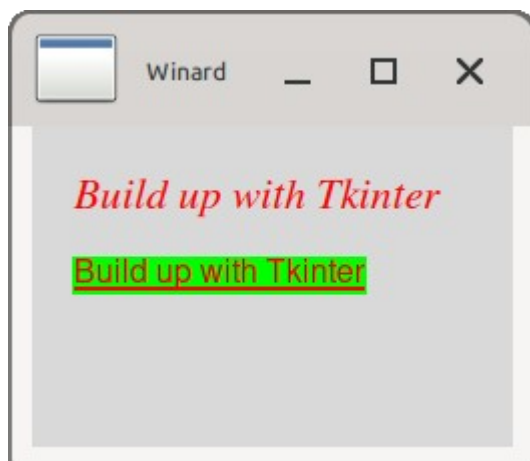
void loop() {}

void build_up()
{ tx_02(PRINTBUILD,1);
  tx_str(TEXT,"Winard");
  tx_04(ROOT,0,240,160);

  tx_str(TEXT,"Build up with Tkinter");
  tx_06(LABEL,3,ROOT,0,20,25);
  tx_10(FONT,0,LABEL,3,1,16,0,1,0,0);
  tx_04(CFG, RED); //color foreground
  tx_05(STYLE,0,LABEL,3,CFG);

  tx_06(LABEL,4,ROOT,0,20,65);
  tx_10(FONT,0,LABEL,4,0,12,0,0,1,0);
  tx_05(STYLE,0,LABEL,4,CFG);
  tx_04(CBG, GREEN); //color background
  tx_05(STYLE,0,LABEL,4,CBG);

  tx_02(PRINTOBJECTS,1);
}
```



The following statements are also valid:

`tx_10(FONT,0,BUTTON,6,0,16,0,1,0,0);` only button 6

`tx_10(FONT,0,THEMESTYLE,0,2,8,0,1,0,0);` for all widgets of the theme

`tx_10(FONT,0,SET,0,2,10,0,1,0,0);`

`tx_05(THEMESTYLE,0,BUTTON,0,FONT);` all buttons of the theme

BUTTON is an example, also valid for RBUTTON , CBUTTON , etc.

Change the theme with:

`tx_02(THEME,*);`

where:

`* = 1 ; clam`

`* = 2 ; alt`

`* = 3 ; default`

`* = 4 ; classic`

4.3 Buttons style

4.3.1 button, radiobutton and checkbutton

```
#include <winard.h>

void setup( )
{ Serial.begin(115200);
  build_up();
}

void loop( ) {}

void build_up()
{ tx_02(PRINTBUILD,1);
  tx_str(TEXT,"Winard");
  tx_04(ROOT,0,240,160);

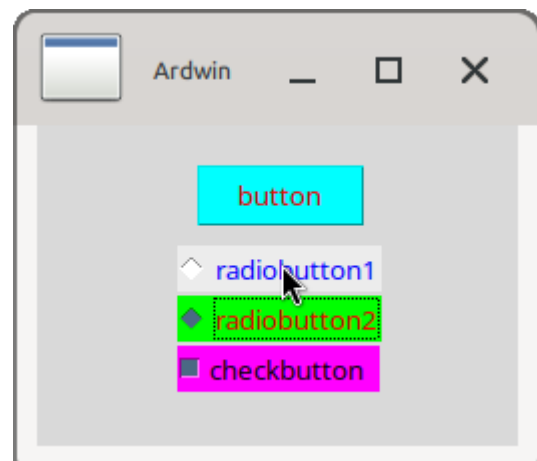
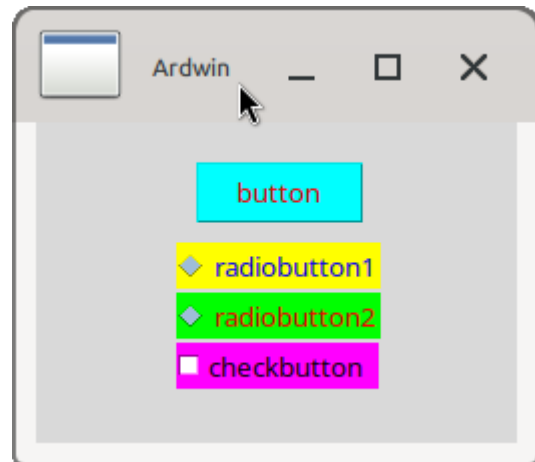
  tx_str(TEXT,"button");
  tx_06(BUTTON,16,ROOT,0,80,20);
  tx_04(CFG,255,0,0);
  tx_05(STYLE,0,BUTTON,16,CFG);
  tx_04(CBG,0,255,255);
  tx_05(STYLE,0,BUTTON,16,CBG);

  tx_str(TAGS,"rb");
  tx_str(TEXT,"radiobutton1");
  tx_06(RBUTTON,17,ROOT,0,70,60);
  tx_04(CFG,0,0,255);
  tx_05(STYLE,0,RBUTTON,17,CFG);
  tx_04(CBG,255,255,0);
  tx_05(STYLE,0,RBUTTON,17,CBG);

  tx_str(TEXT,"radiobutton2");
  tx_06(RBUTTON,18,ROOT,0,70,85);
  tx_04(CFG,255,0,255);
  tx_05(STYLE,0,RBUTTON,18,CFG);
  tx_04(CBG,0,255,0);
  tx_05(STYLE,0,RBUTTON,18,CBG);

  tx_str(TEXT,"checkboxbutton  ");
  tx_06(CBUTTON,19,ROOT,0,70,110);
  tx_04(CFG,0,255,0);
  tx_05(STYLE,0,CBUTTON,19,CFG);
  tx_04(CBG,255,0,255);
  tx_05(STYLE,0,CBUTTON,19,CBG);

  tx_02(PRINTOBJECTS,1);
}
```



4.3.2 Style map

Exept to change the foreground and background colors, it is het possible to change the selected and the not-selected fore- and backgrounds

```
#include <winard.h>

void setup( )
{ Serial.begin(115200);
  build_up();
}

void loop( ) {}

void build_up()
{ tx_02(PRINTBUILD,1);
  tx_str(TEXT,"Winard");
  tx_04(ROOT,0,240,160);

  tx_str(TEXT,"button");
  tx_06(BUTTON,16,ROOT,0,80,20);
  tx_str(TAGS,"rb");
  tx_str(TEXT,"radiobutton1");
  tx_06(RBUTTON,17,ROOT,0,70,60);
  tx_str(TEXT,"radiobutton2");
  tx_06(RBUTTON,18,ROOT,0,70,85);
  tx_str(TEXT," checkboxbutton ");
  tx_06(CBUTTON,19,ROOT,0,70,110);

  tx_04(CFG, WHITE);
  tx_04(CBG, BLACK);
  tx_05(STYLE,0,BUTTON,16,CFG);
  tx_05(STYLE,0,BUTTON,16,CBG);

  tx_04(CACTIVE_FG,ORANGE);
  tx_04(CSELECT, RED);
  tx_04(CNSELECT, YELLOW);

  tx_05(MAP,0, BUTTON,16,CFG);
  tx_05(MAP,0,RBUTTON,17,CFG);
  tx_05(MAP,0,RBUTTON,18,CFG);
  tx_05(MAP,0,CBUTTON,19,CFG);

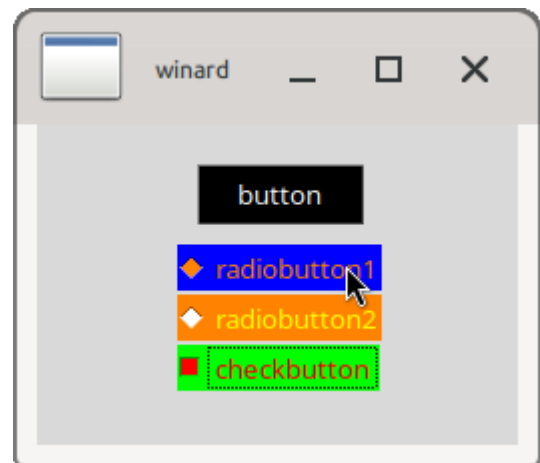
  tx_04(CACTIVE_FG, BLUE);
  tx_04(CSELECT, GREEN);
  tx_04(CNSELECT, ORANGE);

  tx_05(MAP,0, BUTTON,16,CBG);
  tx_05(MAP,0,RBUTTON,17,CBG);
  tx_05(MAP,0,RBUTTON,18,CBG);
  tx_05(MAP,0,CBUTTON,19,CBG);

  tx_04(CACTIVE_FG,MAGENTA);
  tx_04(CSELECT, RED);
  tx_04(CINDICATOR,WHITE);

  tx_05(MAP,0,RBUTTON,17,CIND);
  tx_05(MAP,0,RBUTTON,18,CIND);
  tx_05(MAP,0,CBUTTON,19,CIND);

  tx_02(PRINTOBJECTS,1);
}
```



4.3.3 Button size



```
#include <winard.h>

void setup( )
{ Serial.begin(115200);
  build_up();
}

void loop( ) {;}

void build_up()
{ tx_02(PRINTBUILD,1);
  tx_str(TEXT,"Winard");
  tx_04(ROOT,0,320,240);

  tx_str(TEXT,"text");
  tx_06(BUTTON,3,ROOT,0,10,20);
  tx_06(BUTTON,4,ROOT,0,10,75);
  tx_06(BUTTON,5,ROOT,0,10,130);

  tx_06(BUTTON,13,ROOT,0,125,20);
  tx_06(BUTTON,14,ROOT,0,125,75);
  tx_06(BUTTON,15,ROOT,0,125,130);

  tx_06(BUTTON,23,ROOT,0,240,20);
  tx_06(BUTTON,24,ROOT,0,240,75);
  tx_06(BUTTON,25,ROOT,0,240,130);

  tx_02(BORDERWIDTH,3);
  tx_05(STYLE,0,BUTTON,4,BORDERWIDTH);
  tx_02(BORDERWIDTH,5);
  tx_05(STYLE,0,BUTTON,5,BORDERWIDTH);

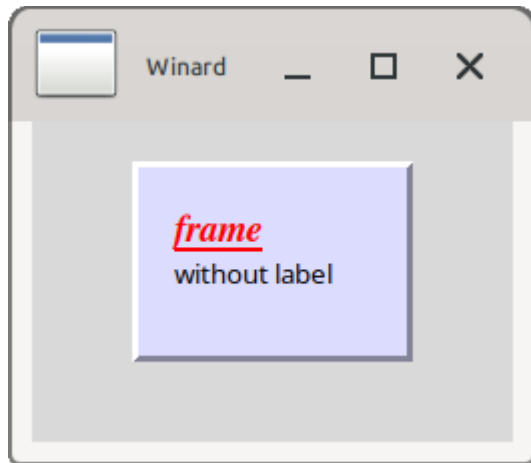
  tx_02(BORDERWIDTH,0);
  tx_05(STYLE,0,BUTTON,13,BORDERWIDTH);
  tx_02(BORDERWIDTH,3);
  tx_05(STYLE,0,BUTTON,14,BORDERWIDTH);
  tx_02(BORDERWIDTH,5);
  tx_05(STYLE,0,BUTTON,15,BORDERWIDTH);

  tx_02(WIDTH,5);
  tx_05(STYLE,0,BUTTON,14,WIDTH);
  tx_02(WIDTH,8);
  tx_05(STYLE,0,BUTTON,15,WIDTH);

  tx_02(WIDTH,2);
  tx_05(STYLE,0,BUTTON,23,WIDTH);
  tx_02(WIDTH,5);
  tx_05(STYLE,0,BUTTON,24,WIDTH);
  tx_02(WIDTH,8);
  tx_05(STYLE,0,BUTTON,25,WIDTH);

  tx_02(PRINTOBJECTS,1);
}
```

4.4 Frame



```
#include <winard.h>

void setup( )
{ Serial.begin(115200);
  build_up();
}

void loop() {};

void build_up()
{ tx_02(PRINTBUILD,1);
  tx_str(TEXT,"Winard");
  tx_04(ROOT,0,240,160);

  tx_str(TEXT,"label");
  tx_02(POSITION,2);
  tx_08(LABELFRAME,1,ROOT,0,30,10,180,130);
  tx_04(CBG, 220,255,220);
  tx_05(STYLE,1,LABELFRAME,1,CBG);
  //      /--- 1 directs to the label
  tx_10(FONT,1,LABELFRAME,1,1,12,0,1,0,0);
  tx_02(RELIEF,2);
  // 0 flat, 1 groove, 2 raised,
  // 3 ridge, 4 solid, 5 sunken
  tx_05(STYLE,0,LABELFRAME,1,RELIEF);
  tx_02(BORDERWIDTH,3);
  tx_05(STYLE,0,LABELFRAME,1,BORDERWIDTH);

  tx_str(TEXT,"frame");
  tx_06(LABEL,4,LABELFRAME,1,40,25);
  tx_str(TEXT,"with label");
  tx_06(LABEL,5,LABELFRAME,1,40,45);
  tx_05(STYLE,0,LABELFRAME,1,CBG);
  tx_05(STYLE,0,LABEL,4,CBG);
  tx_05(STYLE,0,LABEL,5,CBG);

  tx_02(PRINTOBJECTS,1);
}
```

```
#include <winard.h>

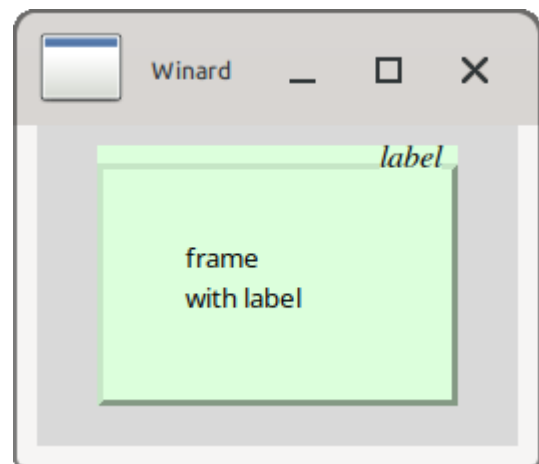
void setup( )
{ Serial.begin(115200);
  build_up();
}

void loop() {};

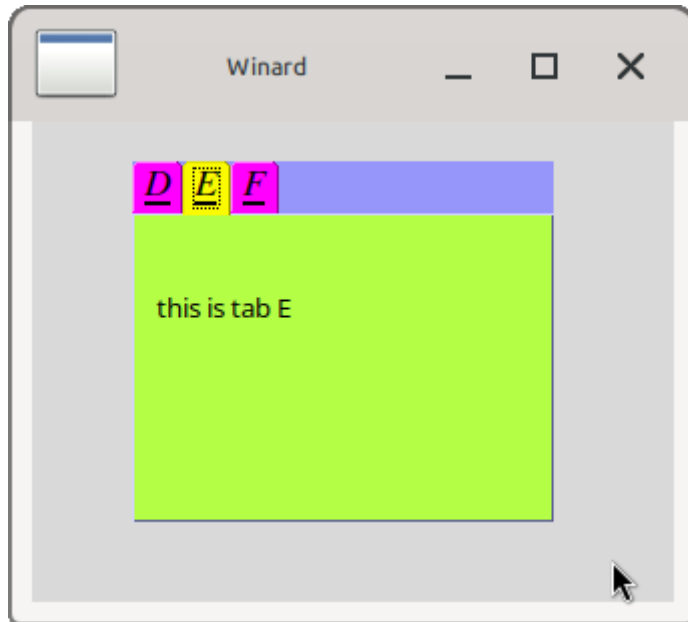
void build_up()
{ tx_02(PRINTBUILD,1);
  tx_str(TEXT,"Winard");
  tx_04(ROOT,0,240,160);

  tx_08(FRAME,1,ROOT,0,50,20,140,100);
  tx_str(TEXT,"frame");
  tx_06(LABEL,4,FRAME,1,20,25);
  tx_04(CFG, 255,0,0);
  tx_05(STYLE,0,LABEL,4,CFG);
  tx_10(FONT,0,LABEL,4,1,14,1,1,1,0);
  tx_str(TEXT,"without label");
  tx_06(LABEL,5,FRAME,1,20,45);
  tx_04(CBG, 220,220,255);
  tx_05(STYLE,0,FRAME,1,CBG);
  tx_05(STYLE,0,LABEL,4,CBG);
  tx_05(STYLE,0,LABEL,5,CBG);
  tx_02(RELIEF,2);
  // 0 flat, 1 groove, 2 raised,
  // 3 ridge, 4 solid, 5 sunken
  tx_05(STYLE,0,FRAME,1,RELIEF);
  tx_02(BORDERWIDTH,3);
  tx_05(STYLE,0,FRAME,1,BORDERWIDTH);

  tx_02(PRINTOBJECTS,1);
}
```



4.5 Notebook



It is not possible to assign a custom numbering to the tabs. It starts with tab0, tab1, tab2 etc.

```
#include <winard.h>

void setup( )
{ Serial.begin(115200);
  build_up();
}

void loop() {};

void build_up()
{ tx_02(PRINTBUILD,1);
  tx_str(TEXT,"Winard");
  tx_04(ROOT,0,320,240);

  // the tabs start with tab 0,
  // the next, tab 1 , tab 2, etc.

  tx_str(TEXT,"D,E,F");
  tx_02(POSITION,0);
  tx_08(NOTEBOOK,2,ROOT,0,50,20,210,180);
  tx_str(TEXT,"this is tab D");
  tx_06(LABEL,14,TAB,0,10,35);
  tx_str(TEXT,"this is tab E");
  tx_06(LABEL,15,TAB,1,10,35);
  tx_str(TEXT,"this is tab F");
  tx_06(LABEL,16,TAB,2,10,35);

  tx_04(CBG,LIME);
  tx_05(STYLE,0,TAB,0,CBG);
  tx_05(STYLE,0,TAB,1,CBG);
  // style the label background too:
  tx_05(STYLE,0,LABEL,15,CBG);

  tx_04(CBG,MAGENTA);
  tx_05(STYLE,1,NOTEBOOK,2,CBG);
  //      \__ directs to tab

  tx_04(CBG,PURPLE);
  tx_05(STYLE,0,NOTEBOOK,2,CBG);
  //      \__ directs to notebook

  tx_04(CSELECT,250,250,0);
  tx_05(MAP,1,NOTEBOOK,2,CSELECT);
  //      \__ directs to tab
  // MAP is a STYLE way too

  tx_10(FONT,0,LABEL,16,1,14,1,0,0,0);
  tx_10(FONT,1,NOTEBOOK,2,1,14,0,1,1,0);
  //      \__ directs to tab
  // FONT is the STYLE way for fonts

  tx_str(TEXT,"ok");
  tx_06(BUTTON,6,TAB,2,40,120);
  tx_02(WIDTH,3); // number of characters
  tx_05(STYLE,1,BUTTON,6,WIDTH);
  tx_04(CBG,150,250,50);
  tx_05(STYLE,1,BUTTON,6,CBG);
  tx_02(PRINTOBJECTS,1);
}
```

4.6 Menubutton

The menu button is in ttk style.

The menu according to tk.

```
#include <winard.h>

void setup( )
{ Serial.begin(115200);
  the_build();
}

void loop() {}

void the_build()
{ tx_02(PRINTBUILD,1);
  tx_str(TEXT,"Winard");
  tx_04(ROOT,0,240,160);

  tx_04(CFG,255,0,0);
  tx_04(CBG,0,255,0);
  tx_04(CSELECT_FG,0,0,255);
  tx_04(CAFG,255,255,0);
  tx_04(CABG,0,255,255);
  tx_10(FONT,0,SET,0,1,12,0,0,1,0);
  tx_08(MENUOPTIONS,1,1,1,1,1,1,1);
  //          1 2 3 4 5 6 7
  // 1-tear 2-font 3-foreground 4-background
  // 5-activeforeground 6-activebackground
  // 7-selectcolor
  tx_str(TEXT,"select");
  tx_06(MBUTTON,2,ROOT,0,80,20);

  tx_04(CBG,220,220,255);
  tx_05(STYLE,0,MBUTTON,2,CBG);
  tx_04(CFG,200,0,0);
  tx_05(STYLE,0,MBUTTON,2,CFG);
  tx_10(FONT,0,MBUTTON,2,1,14,0,1,0,0);

  tx_str(TEXT,"check");
  tx_04(MCBUTTON,3,MBUTTON,2);
  tx_str(TAGS,"rb");
  tx_str(TEXT,"radio1");
  tx_04(MRBUTTON,5,MBUTTON,2);
  tx_str(TEXT,"radio2");
  tx_04(MRBUTTON,6,MBUTTON,2);
  tx_10(FONT,1,MRBUTTON,6,1,14,0,0,1,0);

  tx_02(PRINTOBJECTS,1);
}
```



4.7 Menubar

The menu bar widget is not a themed tk (ttk) functionality.

This means that no style options can be executed afterwards.

All configurations must be present before the widget is created.

This also applies to its components, first define the properties then the component.



```
#include <winard.h>

void setup( )
{ Serial.begin(115200);
  build_up();
}

void loop() {;}

void build_up()
{ tx_02(PRINTBUILD,1);
  tx_str(TEXT,"Winard");
  tx_04(ROOT,0,290,220);

  tx_04(CFG,255,0,0);
  tx_04(CBG,0,255,0);
  tx_04(CSELECT_FG,0,0,255);
  tx_04(CAFG,255,255,0);
  tx_04(CABG,0,255,255);
  tx_10(FONT,0,SET,0,1,13,0,1,1,0);
  tx_08(MENUOPTIONS,1,1,1,1,1,1,1);
  // menubar options: 1 2 3 4 5 6 7
  // 1-tear 2-font 3-foreground 4-background
  // 5-activeforeground 6-activebackground
  // 7-selectcolor

  tx_04(CFG,0,255,0);
  tx_04(CBG,0,0,255);
  tx_04(CSELECT_FG,255,0,0);
  tx_04(CAFG,0,255,255);
  tx_04(CABG,255,0,255);
  tx_10(FONT,0,SET,0,1,11,0,0,0,0);
  tx_08(MENUOPTIONS2,1,1,1,1,1,1,1);
  // barmenu and submenu options

  tx_02(TEAR,1);
  tx_str(TEXT,"menu1,menu2,menu3");
  tx_04(MENUBAR,0,ROOT,0);

  tx_str(TEXT,"button1");
  tx_04(BMCOMMAND,1,BARMENU,0);
  tx_04(BMSEPARATOR,2,BARMENU,0);
  tx_str(TEXT,"submenu1");
  tx_04(CBG,255,0,255);
  tx_08(MENUOPTIONS2,1,1,1,1,1,1,1);
  tx_04(SUBMENU,0,BARMENU,0);
  tx_str(TEXT,"button2");
  tx_04(BMCOMMAND,2,SUBMENU,0);
  tx_str(TEXT,"button3");
  tx_04(BMCOMMAND,3,SUBMENU,0);
  tx_str(TEXT,"submenu2");
  tx_10(FONT,0,SET,0,1,11,0,1,0,0);
  tx_04(CFG,0,0,0);
  tx_04(CBG,255,255,0);
  tx_08(MENUOPTIONS2,1,1,1,1,1,1,1);
  tx_04(SUBMENU,2,SUBMENU,0);
  tx_str(TEXT,"button4");
  tx_04(BMCOMMAND,4,SUBMENU,2);

  tx_str(TAGS,"rb2");
  //TAGS, connects the radio buttons
  // with this label
  tx_str(TEXT,"radio1");
  tx_04(BMRADIO,1,BARMENU,0);
  tx_str(TEXT,"radio2");
  tx_04(CBG,0,255,255);
  tx_08(MENUOPTIONS2,1,1,1,1,1,1,1);
  tx_04(BMRADIO,2,BARMENU,0);
  tx_str(TEXT,"check1");
  tx_04(BMCHECK,1,BARMENU,0);

  tx_str(TEXT,"button5");
  tx_04(BMCOMMAND,5,BARMENU,1);
  tx_str(TEXT,"button6");
  tx_04(BMCOMMAND,6,BARMENU,1);

  tx_02(PRINTOBJECTS,1);
}
```

4.8 Slider or scale

```
#include <winard.h>

void setup( )
{ Serial.begin(115200);
  build_up();
}

void loop() {;}

void build_up()
{ tx_02(PRINTBUILD,1);
  tx_str(TEXT,"Winard");
  tx_04(ROOT,0,240,180);

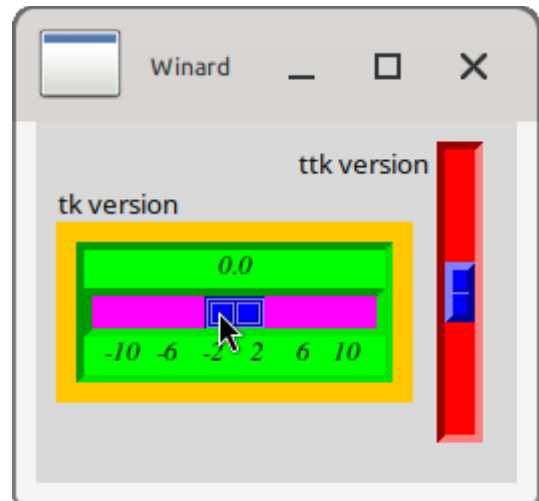
  tx_str(TEXT,"tk version");
  tx_06(LABEL,1,ROOT,0,10,30);

  tx_04(CBG,0,255,0);
  tx_04(CABG,0,0,255);
  tx_04(CTROUGH,255,0,255);
  tx_04(CHLBG,255,200,0); //highlightbackground
  tx_04(CHL,0,200,255);  //highlight
  tx_02(SLIDERWIDTH,16);
  tx_02(RELIEFWIDTH,4);
  tx_02(BORDERWIDTH,10);
  tx_02(RELIEFTROUGH,3);
  tx_02(RELIEFSLIDER,1);
  tx_10(FONT,0,SET,0,1,10,0,1,0,0);
  tx_12(SCALEOPTIONS,1,1,1,1,1,1,1,1,1, 1, 1);
  //      1 2 3 4 5 6 7 8 9 10 11
  //option use, 0 not 1 yes:
  // 1-font 2-background 3-activebackground
  // 4-troughcolor 5-highlightcolor
  // 6-highlightbackground 7-sliderwidth
  // 8-reliefwidth 9-borderwidth
  // 10-relief 11-sliderrelief
  tx_str(TEXT,"0 1 4.0 -10.0 10.0 2.0 1");
  tx_08(SCALE,18,ROOT,0,10,50,150,12);

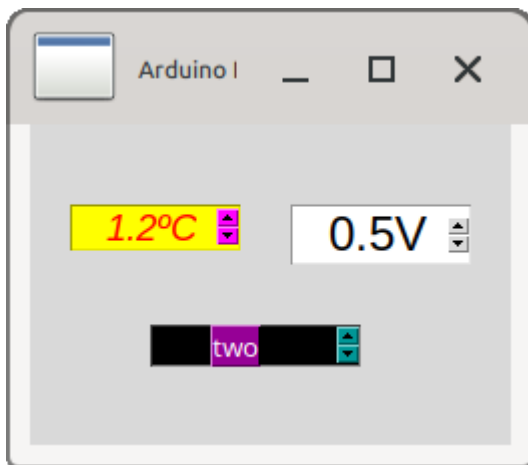
  tx_str(TEXT,"ttk version");
  tx_06(LABEL,1,ROOT,0,130,10);
  tx_str(TEXT,"1 1 2.0 20.0 -20.0 2.0 2");
  tx_08(SCALE,19,ROOT,0,200,10,0,150);

  tx_04(CSLIDER,0,0,255);
  tx_05(STYLE,0,SCALE,19,CSLIDER);
  tx_04(CTROUGH,255,0,0);
  tx_05(STYLE,0,SCALE,19,CTROUGH);
  tx_02(SLIDERWIDTH,15);
  tx_05(STYLE,0,SCALE,19,SLIDERWIDTH);
  tx_02(BORDERWIDTH,4);
  tx_05(STYLE,0,SCALE,19,BORDERWIDTH);
  tx_02(RELIEFTROUGH,1);
  tx_05(STYLE,0,SCALE,19,RELIEFTROUGH);

  tx_02(PRINTOBJECTS,1);
}
```



4.9 Spinbox



```
#include <winard.h>

void setup( )
{ Serial.begin(115200);
  build_up();
}

void loop() {};

void build_up()
{ tx_02(PRINTBUILD,1);
  tx_str(TEXT,"Arduino Display");
  tx_04(ROOT,0,240,160);

  tx_str(SPINBOXSET,"0 2 6 1.0°C");
  tx_str(TEXT,"0.0 10.0 0.1 %.1f°C ");
  tx_10(FONT,0,SPINBOX,1,2,14,0,1,0,0);
  tx_06(SPINBOX,1,ROOT,0,20,40);

  tx_04(CFG, 255,0,0);
  tx_04(CBG, 255,0,255);
  tx_04(CSELECT_FG,0,0,255);
  tx_04(CSELECT_BG,255,100,0);
  tx_04(CFIELD_BG, 255,255,0);
  tx_05(STYLE,0,SPINBOX,1,CFG);
  tx_05(STYLE,0,SPINBOX,1,CBG);
  tx_05(STYLE,0,SPINBOX,1,CFIELD_BG);
  tx_05(STYLE,0,SPINBOX,1,CSELECT_BG);
  tx_05(STYLE,0,SPINBOX,1,CSELECT_FG);

  // remember, next line not longer then the
  // buffer length, including TEXT, spaces
  // and the newline/feedback characters
  tx_str(TEXT,"-0.2V -0.1V 0.0V 0.1V 0.2V 0.5V 1.0V");
  tx_str(SPINBOXSET,"1 2 5 0.2V");
  tx_10(FONT,0,SPINBOX,2,2,18,0,0,0,0);
  tx_06(SPINBOX,2,ROOT,0,130,40);

  tx_str(SPINBOXSET,"1 1 10 four");
  tx_str(TEXT,"one two three four five");
  tx_06(SPINBOX,3,ROOT,0,60,100);

  tx_04(CFG, 255,200,255);
  tx_04(CBG, 0,150,150);
  tx_04(CFIELD_BG,0,0,0);
  tx_04(CSELECT_FG,255,255,255);
  tx_04(CSELECT_BG,150,0,150);
  tx_05(STYLE,0,SPINBOX,3,CFG);
  tx_05(STYLE,0,SPINBOX,3,CBG);
  tx_05(STYLE,0,SPINBOX,3,CFIELD_BG);
  tx_05(STYLE,0,SPINBOX,3,CSELECT_FG);
  tx_05(STYLE,0,SPINBOX,3,CSELECT_BG);
}
```

4.10 Entry

```
#include <winard.h>

void setup( )
{ Serial.begin(115200);
  build_up();
}

void loop() {};

void build_up()
{ tx_02(PRINTBUILD,1);
  tx_str(TEXT,"Winard");
  tx_04(ROOT,0,240,160);

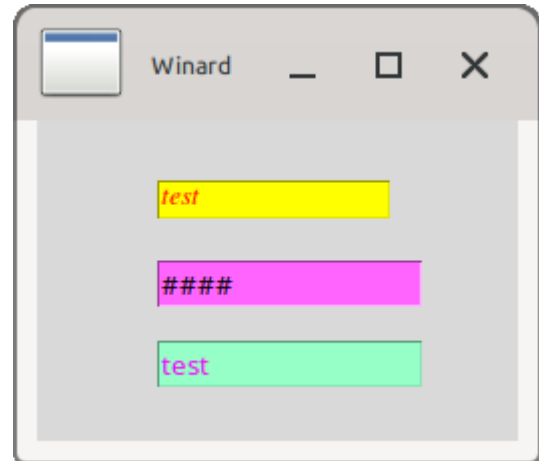
  tx_02(WIDTH,16);
  tx_10(FONT,0,ENTRY,3,1,10,0,1,0,0);
  tx_06(ENTRY,3,ROOT,0,60,30);

  tx_04(CFG, 255,0,0);
  tx_04(CFIELD_BG,255,255,0);
  tx_05(STYLE,0,ENTRY,3,CFG);
  tx_05(STYLE,0,ENTRY,3,CFIELD_BG);

  tx_str(SHOW,"#");
  tx_06(ENTRY,14,ROOT,0,60,70);
  tx_04(CFG, 0,0,0);
  tx_04(CFIELD_BG,255,100,255);
  tx_05(STYLE,0,ENTRY,14,CFG);
  tx_05(STYLE,0,ENTRY,14,CFIELD_BG);

  tx_str(SHOW,"");
  tx_06(ENTRY,23,ROOT,0,60,110);
  tx_04(CFG,255,0,255);
  tx_04(CFIELD_BG,150,255,200);
  tx_05(STYLE,0,ENTRY,23,CFG);
  tx_05(STYLE,0,ENTRY,23,CFIELD_BG);

  tx_02(PRINTOBJECTS,1);
}
```



5 Example's

5.1 Switch leds on-off

```
#include <winard.h>

int red=5, green=6, blue=7;

void setup( )
{ Serial.begin(115200);
  build_up();
  Serial.setTimeout(200);
  pinMode(5, OUTPUT);
  pinMode(6, OUTPUT);
  pinMode(7, OUTPUT);
}

void loop( )
{ readcode();
  if (Bag.i2==27)
  { if (Bag.i3==1)
    { if (Bag.i4==1) digitalWrite(red, HIGH);
      if (Bag.i4==0) digitalWrite(red, LOW);
    }
    if (Bag.i3==2)
    { if (Bag.i4==1) digitalWrite(green,HIGH);
      if (Bag.i4==0) digitalWrite(green,LOW);
    }
    if (Bag.i3==3)
    { if (Bag.i4==1) digitalWrite(blue, HIGH);
      if (Bag.i4==0) digitalWrite(blue, LOW);
    }
  }
}

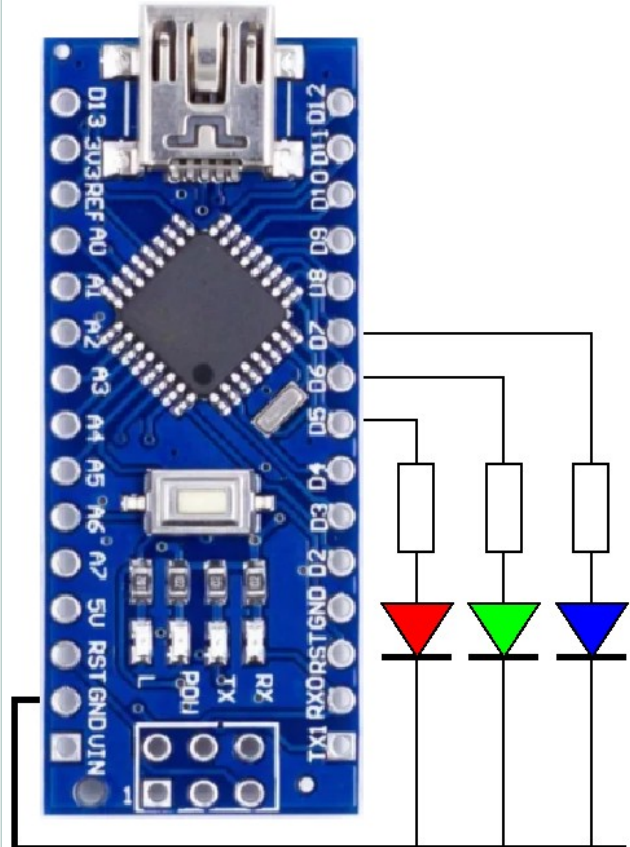
void build_up()
{ tx_02(PRINTBUILD,1);
  tx_str(TEXT,"Winard");
  tx_04(ROOT,0,240,40);

  tx_str(TEXT,"red  ");
  tx_06(CBUTTON,1,ROOT,0,10,10);
  tx_04(CFG,0,0,0);
  tx_04(CBG,255,0,0);
  tx_05(STYLE,0,CBUTTON,1,CFG);
  tx_05(STYLE,0,CBUTTON,1,CBG);

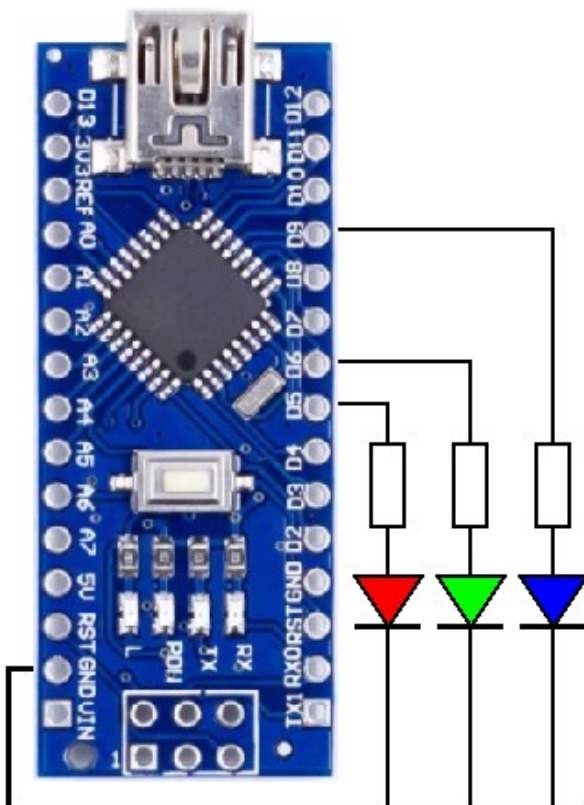
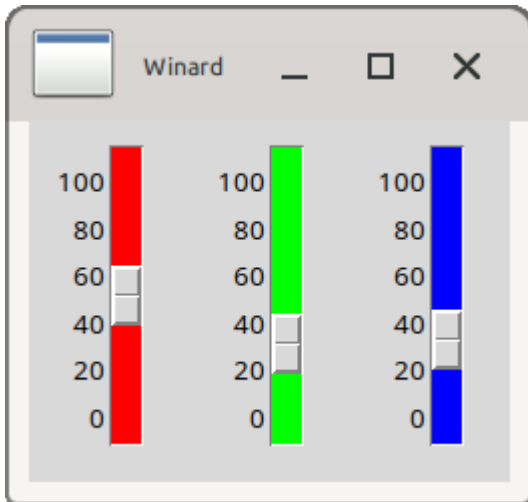
  tx_str(TEXT,"green");
  tx_06(CBUTTON,2,ROOT,0,95,10);
  tx_04(CBG,0,255,0);
  tx_05(STYLE,0,CBUTTON,2,CFG);
  tx_05(STYLE,0,CBUTTON,2,CBG);

  tx_str(TEXT,"blue ");
  tx_06(CBUTTON,3,ROOT,0,180,10);
  tx_04(CFG,255,255,255);
  tx_04(CBG,0,0,255);
  tx_05(STYLE,0,CBUTTON,3,CFG);
  tx_05(STYLE,0,CBUTTON,3,CBG);

  tx_02(PRINTOBJECTS,1);
  // tx_02(PRINTBUILD,0);
}
```



5.2 Led brightness with slider



```
// arduino nano

#include <winard.h>

const int red  =5, green=6, blue =9;

void setup( )
{ Serial.begin(115200);
  build_up();
  Serial.setTimeout(200); // reaction speed
  pinMode(5, OUTPUT);
  pinMode(6, OUTPUT);
  pinMode(9, OUTPUT);
}

void loop( )
{ readcode();
  if (Bag.i2==28)
  { switch(Bag.i3)
    { case 1: analogWrite(red,2*atoi(Bag.str));
      break;
      case 2: analogWrite(green,2*atoi(Bag.str));
      break;
      case 3: analogWrite(blue,2*atoi(Bag.str));
      break;
    }
  }
}

void build_up()
{ tx_02(PRINTBUILD,1);
  tx_str(TEXT,"Winard");
  tx_04(ROOT,0,240,180);

  tx_str(TEXT,"0 0 20 100 0 0 0");
  tx_04(CTROUGH,255,0,0);
  tx_12(SCALEOPTIONS,0,0,0,1,0,0,0,0,0,0);
  tx_08(SCALE,1,ROOT,0,10,10,15,150);

  tx_str(TEXT,"0 0 20 100 0 0 0");
  tx_04(CTROUGH,0,255,0);
  tx_12(SCALEOPTIONS,0,0,0,1,0,0,0,0,0,0);
  tx_08(SCALE,2,ROOT,0,90,10,15,150);

  tx_str(TEXT,"0 0 20 100 0 0 0");
  tx_04(CTROUGH,0,0,255);
  tx_12(SCALEOPTIONS,0,0,0,1,0,0,0,0,0,0);
  tx_08(SCALE,3,ROOT,0,170,10,15,150);

  tx_02(PRINTOBJECTS,1);
}
```

5.3 Draw functions

```
#include <winard.h>
int buf[64];

void setup( )
{ Serial.begin(115200);
  Serial.setTimeout(50);
  build_up();
}

void loop()
{ readcode();
  if (Bag.i2==26)
  { switch(Bag.i3)
    { case 1: tx_04(CFG,255,0,0);break;
      case 2: tx_04(CFG,255,255,0);break;
      case 3: tx_04(CFG,0,255,0);break;
      case 4: tx_04(CFG,0,255,255);break;
      case 5: tx_04(CFG,0,0,255);break;
      case 6: tx_04(CFG,255,0,255);break;
      case 7: tx_04(CFG,255,255,255);break;
    }
  }
  if (Bag.i2==25)
  { switch(Bag.i3)
    { case 1: tx_str(TAGS,"func1"); funcsin(); break;
      case 2: tx_str(TAGS,"func2"); funcsquare(); break;
      case 3: tx_str(TAGS,"func3"); functriangle(); break;
      case 4: tx_str(TAGSDEL,"func1"); ; break;
      case 5: tx_str(TAGSDEL,"func2"); ; break;
      case 6: tx_str(TAGSDEL,"func3"); ; break;
    }
  }
}

void funcsin()
{ int a=0;
  tx_04(SETLINE,1,10,6);
  tx_02(COORD,(int) (50.0*sin(PI * a++/15)+70.0));
  while (a<31)
  {tx_02(COORDAPPEND,(int)(50.0*sin(PI * a++/15)+70.0));
  }
  tx_04(NEWLINE,7,CANVAS,1);
}

void funcsquare()
{ tx_04(SETLINE,0,10,6);
  tx_11(COORD,10,100,10,40,55,40,55,100,100,100);
  tx_11(COORDAPPEND,100,40,145,40,145,100,190,100,190,40);
  tx_04(NEWLINE,8,CANVAS,1);
}

void functriangle()
{ tx_04(SETLINE,1,10,45);
  tx_06(COORD,100,40,100,40,100);
  tx_04(NEWLINE,9,CANVAS,1);
}

void build_up()
{ tx_02(PRINTBUILD,1);
  tx_str(TEXT,"Winard");
  tx_04(ROOT,1,320,240);

  tx_02(WIDTH,8);
  tx_05(THEMESTYLE,0,RBUTTON,0,WIDTH);
```

```

tx_str(TAGS,"rb");
tx_str(TEXT,"red      ");
tx_06(RBUTTON,1,ROOT,0,230,20);
tx_04(CBG,255,0,0);
tx_05(STYLE,0,RBUTTON,1,CBG);
tx_str(TEXT,"yellow   ");
tx_06(RBUTTON,2,ROOT,0,230,50);
tx_04(CBG,255,255,0);
tx_05(STYLE,0,RBUTTON,2,CBG);
tx_str(TEXT,"green");
tx_06(RBUTTON,3,ROOT,0,230,80);
tx_04(CBG,0,255,0);
tx_05(STYLE,0,RBUTTON,3,CBG);
tx_str(TEXT,"cyan");
tx_06(RBUTTON,4,ROOT,0,230,110);
tx_04(CBG,0,255,255);
tx_05(STYLE,0,RBUTTON,4,CFG);
tx_05(STYLE,0,RBUTTON,4,CBG);
tx_str(TEXT,"blue  ");
tx_06(RBUTTON,5,ROOT,0,230,140);
tx_04(CFG,255,255,255);
tx_04(CBG,0,0,255);
tx_05(STYLE,0,RBUTTON,5,CFG);
tx_05(STYLE,0,RBUTTON,5,CBG);
tx_str(TEXT,"magenta");
tx_06(RBUTTON,6,ROOT,0,230,170);
tx_04(CBG,255,0,255);
tx_05(STYLE,0,RBUTTON,6,CBG);
tx_str(TEXT,"white");
tx_06(RBUTTON,7,ROOT,0,230,200);
tx_04(CBG,255,255,255);
tx_05(STYLE,0,RBUTTON,7,CBG);

tx_str(TEXT,"sinus");
tx_06(BUTTON,1,ROOT,0,10,160);
tx_str(TEXT,"square");
tx_06(BUTTON,2,ROOT,0,80,160);
tx_str(TEXT,"triangle");
tx_06(BUTTON,3,ROOT,0,150,160);
tx_str(TEXT,"delete");
tx_06(BUTTON,4,ROOT,0,10,190);
tx_str(TEXT,"delete");
tx_06(BUTTON,5,ROOT,0,80,190);
tx_str(TEXT,"delete");
tx_06(BUTTON,6,ROOT,0,150,190);

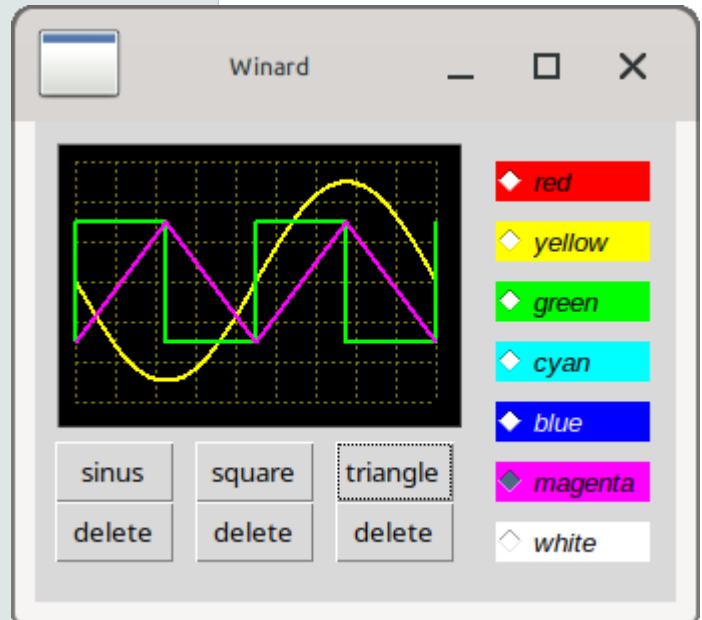
tx_02(WIDTH,6);
tx_05(THEMESTYLE,0,BUTTON,0,WIDTH);

tx_10(FONT,0,SET,0,2,10,0,1,0,0);
tx_05(THEMESTYLE,0,RBUTTON,0,FONT);

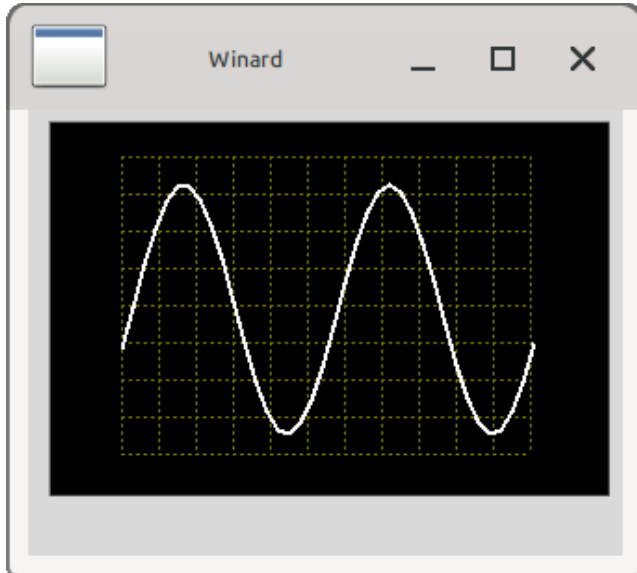
tx_04(CFG,255,255,255);
tx_04(CBG,0,0,0);
tx_08(CANVAS,1,ROOT,0,10,10,200,140);
tx_05(NLINES,7,10,20,20);
tx_04(CFG,120,120,0);
tx_02(LINEWIDTH,1);
tx_04(DASH,1,2,2);
tx_08(VLINES,3,CANVAS,1,10,10,10,130);
tx_08(HLINES,3,CANVAS,1,10,10,190,10);
tx_04(CFG,255,255,255);
tx_04(DASH,0,8,8);
tx_02(LINEWIDTH,2);

tx_02(PRINTOBJECTS,1);
tx_02(PRINTBUILD,0);
}

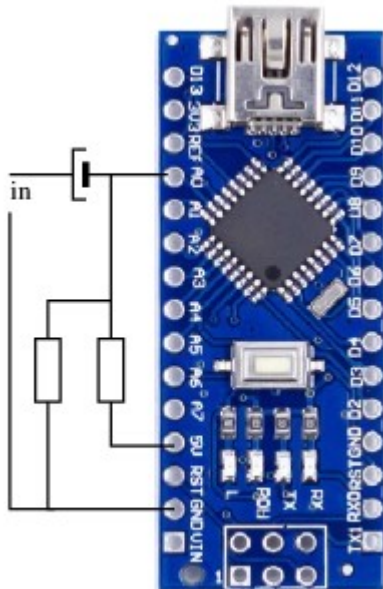
```



5.4 Plot an ADC signal



This signal has a frequency of 48 Hz.



```
//arduino nano
#include <winard.h>

const int input = A0;
int buf[64];

void setup( )
{ Serial.begin(115200);
  build_up();
  // analogReference(INTERNAL);
  // analogReference(EXTERNAL);
  analogReference(DEFAULT);
  Serial.setTimeout(50);
}

void loop()
{ func();
}

void build_up()
{
  tx_02(PRINTBUILD,1);
  tx_str(TEXT,"Winard");
  tx_04(ROOT,1,320,240);

  tx_04(CFG,255,255,255);
  tx_04(CBG,0,0,0);
  tx_08(CANVAS,1,ROOT,0,10,5,300,200);

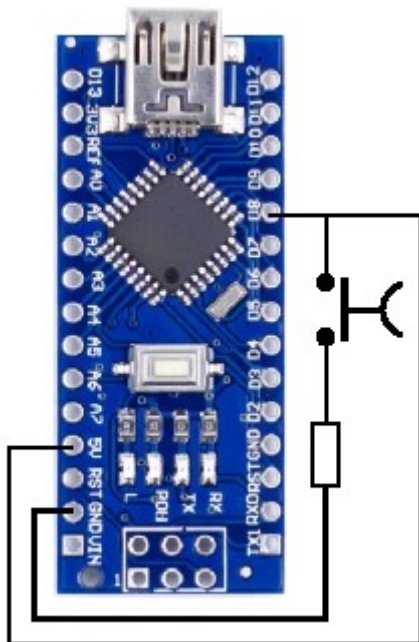
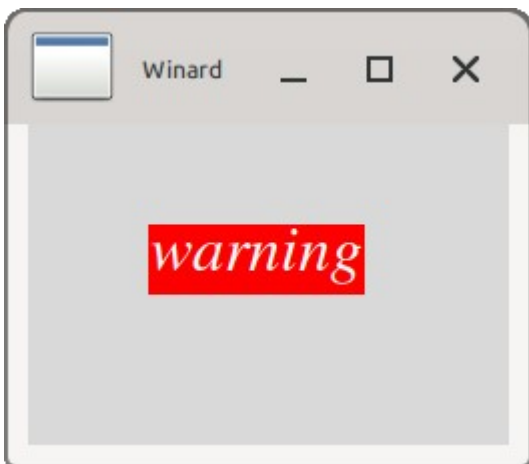
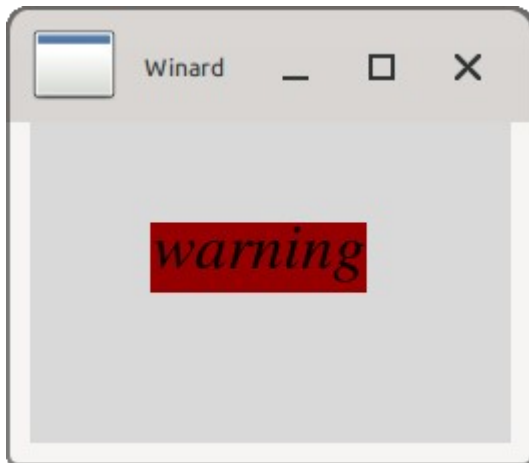
  tx_05(NLINES,9,12,20,20);
  tx_04(CFG,120,120,0);
  tx_02(LINEWIDTH,1);
  tx_04(DASH,1,2,2);
  tx_08(VLINES,3,CANVAS,1,40,20,40,180);
  tx_08(HLINES,3,CANVAS,1,40,20,260,20);
  tx_04(CFG,255,255,255);
  tx_04(DASH,0,8,8);

  tx_04(SETLINE,1,40,6);
  //          | | |_fix.distances X/Y dir.
  //          | |__startvalue X or Y
  //          |__=0 line with X,Y amplitudes
  //          |__=1 line, only Y amplitudes
  //          |__=2 line, only X amplitudes
  tx_str(TAGS,"two");
  tx_02(LINEWIDTH,2);

  tx_02(PRINTOBJECTS,1);
  tx_02(PRINTBUILD,0);
}

void func()
{ int a=0;
  while (a<(38))
  { buf[a++]=analogRead(A0);delay(1);
  }
  a=0;
  tx_02(COORD,buf[a++]/6+15);
  while (a<(38))
  { tx_02(COORDAPPEND,buf[a++]/6+15);delay(2);
  }
  tx_str(TAGSDEL,"two");
  tx_04(NEWLINE,9,CANVAS,1);
}
```

5.5 Signal a warning



```
#include <winard.h>

const int buttonPin = 8;
int buttonState = 0;
int mem=0;

void setup()
{ Serial.begin(115200);
  build_up();
  pinMode(buttonPin, INPUT);
}

void loop()
{ buttonState = digitalRead(buttonPin);
  if ((buttonState == HIGH) & (mem==0))
  { tx_04(CBG, 255,0,0);
    tx_05(STYLE,0,LABEL,3,CBG);
    tx_04(CFG,255,255,255);
    tx_05(STYLE,0,LABEL,3,CFG);
    mem=1;
  }
  if ((buttonState == LOW) & (mem==1))
  { tx_04(CBG, 150,0,0);
    tx_05(STYLE,0,LABEL,3,CBG);
    tx_04(CFG,0,0,0);
    tx_05(STYLE,0,LABEL,3,CFG);
    mem=0;
  }
}

void build_up()
{ tx_02(PRINTBUILD,1);
  tx_str(TEXT,"Winard");
  tx_04(ROOT,0,240,160);

  tx_str(TEXT,"warning");
  tx_06(LABEL,3,ROOT,0,60,50);
  tx_10(FONT,0,LABEL,3,1,24,0,1,0,0);
  tx_04(CFG,0,0,0);
  tx_05(STYLE,0,LABEL,3,CFG);
  tx_04(CBG, 150,0,0);
  tx_05(STYLE,0,LABEL,3,CBG);

  tx_02(PRINTOBJECTS,1);
}
```